

Affine Rigidity of a Named Complement Core

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Abstract

The preceding complement atlas realizes its universal core as $Q \cong \mathbb{Z}/9 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/2$ with sixteen named coordinate occurrences. The preceding deletion paper found an abstract generation-hypergraph automorphism group of order 345600, while leaving its relation to the actual core open. We enumerate the finite affine group of the frozen realization: $|\text{Aut}(Q)| = 108$ and $|\text{Aff}(Q)| = 5832$. The sixteen-occurrence multiset and its ten-point underlying set each have trivial affine stabilizer. We give complete occurrence- and point-overlap censuses; the nearest nonidentity affine maps overlap fourteen of sixteen occurrences. This is a rigidity statement about one named presentation, not a claim that the combinatorial hypergraph is a core-group symmetry.

1 Frozen named realization

Let M be the multiset of the sixteen C72 coordinates in

$$Q = \mathbb{Z}/9 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/2,$$

and let $X = \text{supp}(M)$. Thus $|M| = 16$, $|X| = 10$, and the multiplicity profile of M is

$$5, 2, 2, 1, 1, 1, 1, 1, 1, 1.$$

The two repeated nonzero points are $(0, 1, 0)$ and $(3, 1, 0)$; the point $(0, 0, 0)$ has multiplicity five. We retain the named presentation rather than choosing a new basis. The scope firewall is

NO_BAD_EULER_OR_ROOT_NUMBER.

No full Burnside ring, arithmetic or local construction, Euler factor, root number, automorphy, or Hilbert–Polya operator is claimed.

2 The finite affine group

Put $P = \mathbb{Z}/9 \oplus \mathbb{Z}/3$. Every endomorphism of P has the form

$$(x, y) \mapsto (ax + 3by \bmod 9, cx + dy \bmod 3), \tag{1}$$

where $a \in \mathbb{Z}/9$ and $b, c, d \in \mathbb{Z}/3$. The dyadic factor has only the identity automorphism.

Theorem 1 (Affine-group order). *There are 243 endomorphism matrices of P in (1); adjoining the two endomorphisms of the $\mathbb{Z}/2$ factor gives 486 endomorphisms of Q . A full endomorphism is an automorphism exactly when its dyadic component is the identity and a and d are nonzero modulo 3. Consequently*

$$|\text{Aut}(Q)| = 6 \cdot 3 \cdot 3 \cdot 2 = 108, \quad |\text{Aff}(Q)| = 54 \cdot 108 = 5832.$$

Proof. Modulo $3P$, the induced matrix is triangular with diagonal entries $\bar{a}, d$. This proves the criterion; direct enumeration on all 54 group points gives the same bijective list. The affine group is the semidirect product of translations by Q with the automorphism group. $\square$

3 Rigidity of points and occurrences

For an affine map φ , define the occurrence overlap by

$$O(\varphi) = \sum_{q \in Q} \min(m_M(q), m_{\varphi(M)}(q)). \quad (2)$$

This counts duplicate occurrences. It is distinct from a pointwise fixed-label count and from $|X \cap \varphi(X)|$.

Theorem 2 (Named-core rigidity).

$$\text{Stab}_{\text{Aff}(Q)}(M) = 1, \quad \text{Stab}_{\text{Aff}(Q)}(X) = 1.$$

The corresponding linear stabilizers are also trivial. The linear and affine orbit sizes of either named object are therefore 108 and 5832.

Proof. For M , the unique multiplicity-five point is zero, so a stabilizing translation vanishes. The unique nonzero point of $M \cap 3P$ is $(6, 0, 0)$, which gives $a \equiv 1 \pmod{3}$. The multiplicity-two pair $u = (0, 1, 0)$, $v = (3, 1, 0)$ is preserved setwise. Equation (1) first gives $d = 1$ and $b \in \{0, 1\}$; if $b = 1$, the image of v has first coordinate 6, outside the pair, so $b = 0$. The image of $(1, 0, 0)$ can only be the named order-nine point $(1, 0, 0)$ or $(4, 2, 0)$; the latter would map $(4, 2, 0)$ to the absent point $(7, 1, 0)$. Hence $a = 1, c = 0$.

For X , its unique point with dyadic coordinate one first forces the dyadic translation to vanish. The sum of the ten distinct points is $(0, 0, 1)$; every automorphism fixes the unique order-two element, so preservation of this sum together with the affine sum identity gives $10t = 0$ in the odd factors. Since $10 \equiv 1$ modulo both 9 and 3, the odd translation also vanishes. Reducing the order-nine points modulo three gives $T = \{(1, 0), (1, 2), (2, 1), (2, 2)\}$; checking the six possible induced (c, d) pairs forces $c = 0, d = 1$, and the remaining representatives force $a = 1, b = 0$. Thus both stabilizers are trivial. $\square$

4 Exact overlap census

The complete occurrence-overlap histogram is

O	0	1	2	3	4	5	6	7
count	1435	1339	1068	771	621	265	134	87
O	8	9	10	11	12	13	14	16
count	26	29	35	7	10	2	2	1

Table 1: Occurrence overlap over all 5832 affine maps.

The distinct-point intersection histogram, in contrast, is

$ X \cap \varphi(X) $	0	1	2	3	4	5	6	7	8	10
count	1435	1346	1139	929	628	275	63	14	2	1.

The identity is the unique occurrence overlap 16. The largest nonidentity value is 14, attained exactly by

$$(x, y, z) \mapsto (4x, 2x + y, z), \quad (x, y, z) \mapsto (7x, x + y, z),$$

whose distinct-point overlap is 8.

5 Symmetry boundary and conclusion

C73's minimal-generation hypergraph on sixteen labels has abstract automorphism order

$$6! 2! 2! 5! = 345600.$$

C74 proves that no nonidentity affine map of the actual named core permutes the named point set. These are different structures: the first permutes combinatorial labels while the second acts on points of a fixed finite abelian group. After the point map is fixed, duplicate labels admit the bookkeeping fiber $5! 2! 2! = 480$; this is not an affine stabilizer.

The producer, independent 54-point checker, alternate matrix/image cross-check, clean replay, and hostile mutations certify the census. All claims remain confined to the frozen named presentation.